

1. Core Java Fundamentals

1. What is Java?

Answer: Java is a high-level, object-oriented programming language designed for platform independence via the Java Virtual Machine (JVM).

2. What are JDK, JRE, and JVM?

Answer: JDK (Java Development Kit) is for development, JRE (Java Runtime Environment) to run Java programs, and JVM (Java Virtual Machine) executes Java bytecode.

3. Explain the main features of Java.

Answer: Key features include platform independence, object-orientation, automatic memory management (garbage collection), multi-threading, and robust security.

4. What is the difference between JDK and JRE?

Answer: JDK includes tools for developing Java applications (compiler, debugger), while JRE only contains the JVM and libraries needed to run Java programs.

5. What are primitive data types in Java?

Answer: byte, short, int, long, float, double, char, boolean.

6. What are wrapper classes?

Answer: Classes in `java.lang` that encapsulate primitives as objects (e.g., `Integer` for `int`, `Double` for `double`).

7. What is the difference between `==` and `.equals()`?

Answer: `==` checks reference equality; `.equals()` checks logical/content equality (if overridden).

8. What is type casting?

Answer: Converting one data type to another, either implicitly (widening) or explicitly (narrowing).

9. What is a constant in Java?

Answer: A `final` variable whose value cannot change once assigned.

10. What is a Java package?

Answer: A namespace grouping related classes and interfaces, aiding organization and access control.

2. Object-Oriented Programming (OOP)

1. What are the four pillars of OOP?

Answer: Encapsulation, Inheritance, Polymorphism, Abstraction.

2. What is encapsulation?

Answer: Wrapping data (fields) and methods into a single unit (class), and controlling access via access modifiers.

3. What is inheritance?

Answer: Mechanism where a class (subclass) inherits fields and methods from another class (superclass).

4. What is polymorphism?

Answer: Ability of an object to take many forms, achieved via method overloading (compile-time) and overriding (run-time).

5. What is abstraction?

Answer: Hiding implementation details and exposing only essential features via abstract classes or interfaces.

6. Difference between abstract class and interface.

Answer: Abstract classes can have implemented methods and fields; interfaces (pre-Java 8) only declared methods. Java 8+ interfaces can have default and static methods.

7. What is method overloading?

Answer: Defining multiple methods with the same name but different parameter lists in the same class.

8. What is method overriding?

Answer: Subclass provides a specific implementation for a method declared in its superclass.

9. What is `super` keyword?

Answer: Refers to superclass objects; used to access superclass methods/constructors.

10. What is a constructor?

Answer: Special method to initialize new objects. Java provides a default constructor if none is defined.

3. Collections Framework

1. What is a Collection?

Answer: A framework for storing and processing groups of objects; includes interfaces like `List`, `Set`, and `Map`.

2. Difference between `ArrayList` and `LinkedList`.

Answer: `ArrayList` is backed by a dynamic array (fast random access); `LinkedList` is a doubly-linked list (fast insert/delete).

3. **What is a `HashMap` ?**

Answer: An implementation of `Map` storing key-value pairs with hashing for fast retrieval.

4. **What is a `HashSet` ?**

Answer: An implementation of `Set` that stores unique elements with hashing.

5. **What is `TreeMap` and when to use it?**

Answer: A `Map` implementation that stores keys in sorted order (uses a red-black tree).

6. **What are differences between `List`, `Set`, and `Map` ?**

Answer: `List` is ordered and allows duplicates; `Set` is unordered and disallows duplicates; `Map` stores key-value pairs.

7. **What is an iterator?**

Answer: An object to iterate through a Collection supporting `hasNext()` and `next()`.

8. **How to synchronize a collection?**

Answer: Use `Collections.synchronizedList(list)` or concurrent collections like `CopyOnWriteArrayList`.

9. **What is `ConcurrentHashMap` ?**

Answer: A thread-safe Map implementation that allows concurrent read and limited concurrent write operations.

10. **How to sort a list?**

Answer: Use `Collections.sort(list)` for natural order or `list.sort(Comparator)` for custom order.

4. Exception Handling

1. **What is exception handling?**

Answer: Mechanism to handle runtime errors using `try`, `catch`, `finally`, `throw`, and `throws`.

2. **Difference between checked and unchecked exceptions.**

Answer: Checked exceptions must be declared/handled (e.g., `IOException`); unchecked are runtime exceptions (e.g., `NullPointerException`).

3. **What is `finally` block?**

Answer: Code in `finally` always executes after `try` / `catch`, used for cleanup.

4. **What is `throw` and `throws` ?**

Answer: `throw` manually throws an exception; `throws` declares that a method may throw exceptions.

5. **How to create custom exceptions?**

Answer: Extend `Exception` (checked) or `RuntimeException` (unchecked) in a new class.

5. Multithreading & Concurrency

1. How to create a thread?

Answer: Implement `Runnable` or extend `Thread` and call `start()`.

2. What is synchronization?

Answer: Ensuring only one thread accesses a block/method at a time using `synchronized` keyword.

3. What is a deadlock?

Answer: Situation where two or more threads wait indefinitely for locks held by each other.

4. What are thread states?

Answer: NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED.

5. What is `ExecutorService`?

Answer: Framework for managing thread pools and asynchronous tasks.

6. JDBC & Database Connectivity

1. What is JDBC?

Answer: Java Database Connectivity API for interacting with relational databases.

2. How to connect to a database?

Answer: Load driver, call `DriverManager.getConnection()`, then create `Statement` / `PreparedStatement`.

3. Difference between `Statement` and `PreparedStatement`.

Answer: `PreparedStatement` precompiles SQL and allows safe parameter binding (prevents SQL injection).

4. How to handle transactions?

Answer: Use `conn.setAutoCommit(false)`, then `conn.commit()` or `conn.rollback()`.

5. What is connection pooling?

Answer: Reusing database connections via a pool to improve performance and resource usage.

7. Java 8 Features

1. What are lambda expressions?

Answer: Anonymous functions that implement functional interfaces, enabling concise code.

2. What are Streams?

Answer: API for functional-style processing of collections (e.g., filter, map, reduce).

3. **Difference between** `map()` **and** `flatMap()`.

Answer: `map()` transforms each element; `flatMap()` transforms and flattens nested streams.

4. **What is** `Optional`?

Answer: Container that may or may not hold a non-null value, avoiding null checks.

5. **What is the new Date/Time API?**

Answer: `java.time` package (`LocalDate`, `LocalTime`, `LocalDateTime`) providing immutable and thread-safe date/time handling.

Note: Each question and answer above is designed for quick review and foundational understanding. Freshers should supplement this with hands-on practice and deeper study.